Audit Before Accuracy: Explainable Ordinal Deep Learning for Androgenetic Alopecia Severity Grading on a Leakage-Corrected Benchmark

Mahibi Islam
Department of Computer Science and Engineering
BRAC University, Dhaka, Bangladesh

***Abstract*—Androgenetic alopecia (AGA) affects up to half of men by age 50 and is staged clinically on the seven-point Norwood–Hamilton scale, an *ordinal* scale whose adjacent stages overlap so strongly that expert inter-observer reliability is only ICC 0.63–0.68. Existing deep-learning work treats staging as flat multi-class classification, penalising a one-stage error identically to a five-stage error, and reports accuracy on public benchmarks whose integrity is rarely verified. We address both gaps. First, we audit the widely used `hair_norwood_hamilton`benchmark and show it is unusable as published: its 5,783 files resolve to only 1,447 independent photographs, 53.2% of its test split has a perceptual near-duplicate in train, and two non-clinical artefacts—the vendor's own colour augmentation and the source collection—are strongly confounded with the label. A control model using *28 colour statistics only* reaches QWK 0.722 on this split, establishing a shortcut floor that any reported number must clear. Second, on a leakage-corrected, group-disjoint protocol we compare a softmax head against the CORAL and CORN ordinal formulations on a shared ResNet-50 trunk. CORN is best on all six primary metrics (QWK 0.900, MAE 0.283 stages, macro-F1 0.580), improving on softmax by $+0.031$ QWK and -0.056 MAE, while CORAL exhibits a previously unreported posterior pathology (48.3% of its point predictions fall in a low-mass valley; ECE 0.300). On a SHA-256-locked test set (n=289) the final model attains QWK 0.891 [0.832, 0.927] with 97.6% of predictions within one stage. A controlled ablation—identical architecture, recipe and seed—shows the unaudited split inflates accuracy by 15.1 points and macro-F1 by 0.253. We release the audit, the locked split manifest and a model card.**

***Index Terms*—androgenetic alopecia, ordinal regression, data leakage, shortcut learning, explainable AI, Grad-CAM, medical image grading, model calibration**

I. Introduction

Androgenetic alopecia (AGA) is the most common form of hair loss, affecting up to 50% of males by age 50 [1]. It is progressive and androgen-driven, causing miniaturisation of terminal scalp hair, and carries a documented psychological burden. The clinical standard for staging severity is the seven-point Norwood–Hamilton scale [2], [3], which grades hairline recession and vertex thinning.

Two properties of this problem are routinely mishandled by automated systems.

The scale is ordinal, not categorical. Predicting Stage 3 for a Stage 4 patient is a different clinical event from predicting Stage 1 for a Stage 6 patient. Flat softmax classification assigns both the same loss and the same error count. Since treatment pathways differ sharply between early and advanced stages, a metric that cannot distinguish an adjacent confusion from a gross one misrepresents clinical risk.

The scale is unreliable even among experts. Guarrera *et al.* [4] had seven dermatologists and sixteen residents stage 43 photographs and found concordance of only ICC 0.63–0.68, with poor intra-observer repeatability on re-test. Any model claiming near-perfect agreement with these labels is, on its face, agreeing with something other than the clinical construct.

A third problem is methodological rather than clinical. Public computer-vision benchmarks are often re-exports of scraped collections, and the pre-processing applied before export can leave label-correlated artefacts behind. In medical imaging this has repeatedly produced published results that do not survive an integrity check [5], [6]. We find that the most widely available public Norwood–Hamilton dataset has exactly this problem, and that the problem is large enough to dominate any architectural comparison performed on it.

A. Contributions

1) **A reproducible audit** of the public `hair_norwood_hamilton` benchmark, identifying four independent defects: 53.2% test-set leakage invisible to byte-level deduplication, a label-confounded augmentation artefact, a label-confounded source collection, and an effective sample size $4\times$ smaller than the file count implies (Section IV-A).
2) **A shortcut floor.** A logistic regression on 28 global colour statistics—which cannot resolve hair at all—reaches QWK 0.722 and 82.3% on "Stage $\leq$2 vs. $\geq$3" on the published split. We argue this is the minimum bar for any claim on this benchmark.
3) **A leakage-corrected protocol**: perceptual-hash grouping with union-find, one image per independent group, and a SHA-256-locked test set, released as a manifest.
4) **A controlled ordinal comparison** of softmax, CORAL [7] and CORN [8] heads on a shared trunk, with order-aware metrics and bootstrap intervals.
5) **A quantified leakage ablation**: the same model, recipe and seed evaluated under both protocols, isolating how much the unaudited benchmark inflates each reported metric.

We frame the outcome explicitly as a research prototype, not a diagnostic device.

II. Background and Related Work

A. AGA Staging and Its Reliability

The Hamilton [2] and Norwood [3] classifications define seven ordered stages from minimal recession (I) to complete vertex and frontal loss (VII). The scale is practical and near-universal in trichology, but it is defined on qualitative visual patterns and its boundaries are not sharp. Beyond the ICC 0.63–0.68 of [4], subsequent work has proposed adapted and region-specific alternatives precisely because adjacent-stage discrimination is the dominant source of disagreement. This establishes two things for our study: an upper bound on achievable exact agreement, and a reason to treat adjacent confusion as qualitatively different from distant confusion.

B. Deep Learning for AGA Severity

Recent work applies convolutional networks to trichoscopic and scalp photography. Gao *et al.* [9] built a quantitative model over trichoscopic images; Di Fraia *et al.* [10] applied classical machine learning to trichoscopy; Suh *et al.* [11] report automated early detection on a Korean cohort; Xi *et al.* [12] stratify male-pattern hair loss using a loss-region-ratio analysis. Reported accuracies in this literature commonly exceed 90%. Three characteristics recur: severity is framed as flat classification or binarised; evaluation is by accuracy and unweighted metrics; and splitting is performed at the image level, with patient- or photograph-level independence rarely demonstrated. Our audit suggests the third characteristic is not a minor bookkeeping detail.

C. Ordinal Regression in Deep Networks

Niu *et al.* [13] decomposed ordinal prediction into $K-1$ binary "is $y > k$?" subproblems (OR-CNN), but the independent binary heads can produce rank-inconsistent outputs, i.e. $\hat{P}(y > 2) > \hat{P}(y > 1)$. CORAL [7] removes the inconsistency architecturally: all $K-1$ classifiers share a single weight vector and differ only in bias, so monotonicity holds whenever the learned biases are ordered. The cost is expressiveness—one direction in feature space must serve every threshold. CORN [8] achieves the same guarantee without the constraint, by having head j estimate a *conditional* probability trained only on the subset of samples that have already passed threshold $j-1$; unconditional probabilities are then reconstructed as a running product, which is monotone by construction. We evaluate both, since the proposal's "ordinal cumulative-threshold formulation" admits either.

D. Leakage and Shortcut Learning

Geirhos *et al.* [6] characterise shortcut learning: networks preferentially exploit superficial cues that correlate with the label on the training distribution. Roberts *et al.* [5] reviewed several hundred COVID-19 imaging models and found essentially none clinically usable, with duplicated images and improper splitting among the dominant failure modes. Kaufman *et al.* [14] formalise leakage as the introduction of information about the target that would not be legitimately available at inference time. Our audit instantiates all three concerns simultaneously on a single public benchmark, and —unusually—measures the size of the resulting inflation directly.

III. Methodology

Our pipeline follows a strict order: *audit first, train second, evaluate once*.

A. Dataset

We use `hair_norwood_hamilton`from Roboflow Universe (CC BY 4.0) [15], exported in multi-class classification format: 5,783 JPEG images at 640×640, each labelled with exactly one of the seven Norwood stages, pre-split by the vendor into 5,029 train / 506 validation / 248 test. All images are top-down or vertex-oblique scalp photographs of variable quality, acquired under uncontrolled conditions.

B. Audit Protocol

Integrity. We rebuild a manifest from the per-split label CSVs and verify that every row has exactly one positive class, that no file is missing, and that no image on disk is unlisted. All 5,783 rows pass; there are no unlabelled or multi-label rows.

Exact duplicates. MD5 hashing of file bytes returns zero duplicate groups. This is the expected result—the export platform deduplicates by content hash—and is precisely why byte-level checking is insufficient here.

Near-duplicates. Filenames follow the pattern

```
<source_id>[_<aug>]_jpg.rf.<hash>.jpg
```

where `<source_id>` identifies the original photograph and the optional `<aug>` token marks a colour variant baked in before upload. We compute a 64-bit DCT perceptual hash for every image and link any pair with Hamming distance $\leq \tau$. Independent groups are then the connected components of the union of (i) the same-`source_id` relation and (ii) the near-duplicate relation, computed with union-find.

Threshold selection. We sweep τ. Group count is flat at $\approx$1,449 for $\tau \in \{0, 3\}$ and 1,447 at τ=5, then collapses: at τ=10 a single component absorbs 581 images, i.e. distinct subjects are being merged. We adopt τ=5, inside the stable plateau. For reference, randomly paired images in this corpus differ by 29.5 ± 5.5 bits, so τ=5 is a conservative choice.

Confound analysis. We cross-tabulate the augmentation token and the source-collection family against the stage label and report χ^2 with Cramér's V.

Colour-only control. We extract 28 global colour statistics per image (mean and standard deviation of RGB and HSV, plus 8-bin hue and saturation histograms) from a 32×32 thumbnail and fit a class-balanced multinomial logistic regression. These features carry no shape, texture or scalp-structure information. Their score is our shortcut floor.

C. Canonical Dataset and Splits

We retain one representative image per independent group, preferring an un-augmented original and breaking ties deterministically by filename. Group labels are assigned by majority vote across the group's members; six groups whose own copies

carry conflicting labels are excluded as ambiguous, leaving $n = 1{,}441$ images. We then draw a stratified 20% locked test set (n=289), never touched until final evaluation, and run 5-fold stratified cross-validation on the remaining 1,152 development images, benchmarking over 3 folds. Because the canonical dataset already contains one image per group, group-disjointness is guaranteed by construction; we verify it explicitly. The split manifest is written to disk and fingerprinted with SHA-256 (`dc94878c9541...`), making the "locked test set" claim checkable rather than asserted.

D. Models

All deep models share an ImageNet-pretrained ResNet-50 [16], [17] trunk so that the comparison isolates the output head. Inputs are 224×224 RGB with standard ImageNet normalisation. We attach global average pooling, dropout (p=0.35), and one of three heads.

Softmax (CE). K outputs with categorical cross-entropy and label smoothing 0.05.

CORAL [7]. A single shared projection $g(\mathbf{x}) = \mathbf{w}^\top f(\mathbf{x})$ plus $K-1$ independent bias terms b_j, trained with unconditional binary targets:

$$\mathcal{L}_{\text{CORAL}} = \sum_{j=1}^{K-1} \ell_{\text{BCE}}(\sigma(g(\mathbf{x}) + b_j),\, \mathbb{1}[y > j])\,. \qquad (1)$$

CORN [8]. $K-1$ independent logits. Head j estimates the conditional probability $f_j(\mathbf{x}) = \hat{P}(y > j \mid y > j-1)$ and is trained only on the conditional subset $S_j = \{i : y_i \geq j\}$:

$$\mathcal{L}_{\text{CORN}} = \sum_{j=1}^{K-1} \mathbb{1}[\, i \in S_j \,]\; \ell_{\text{BCE}}(\sigma(z_j),\, \mathbb{1}[y_i > j])\,. \qquad (2)$$

Unconditional probabilities are reconstructed as a running product,

$$\hat{P}(y > k) \;=\; \prod_{j \leq k} f_j(\mathbf{x}), \qquad (3)$$

which is monotonically non-increasing in k because every factor lies in $[0, 1]$—rank consistency follows from the chain rule rather than from a weight constraint. We emit a per-sample loss so that class-balancing sample weights compose correctly; a consequence is that a Stage-VII example contributes $K-1$ active terms while a Stage-I example contributes one, which mildly upweights the rare advanced stages.

Decoding. From $\hat{P}(y > k)$ we obtain class posteriors by differencing the padded cumulative vector, and predict by the expected-value rule $\hat{y} = \text{round}\Big(\sum_k \hat{P}(y > k)\Big)$, which is better matched to QWK and MAE than the arg-max rule.

Baselines. Majority class; the colour-only logistic regression; an RBF-SVM and an ordinal (cumulative binary) SVM on frozen ResNet-50 features; and a four-block CNN trained from scratch. We also evaluate an EfficientNet-B0 [18] trunk with a CORN head.

E. Training

Two-stage transfer learning: (A) frozen trunk, head only, Adam at 10^{-3} for up to 15 epochs; (B) fine-tuning from `conv4_block1_1_conv` onward at 10^{-5} for up to 35 epochs with BatchNorm statistics frozen, which is standard practice for small medical corpora. Early stopping and checkpointing monitor *validation QWK*, not validation accuracy—selecting on accuracy under a 37:1 imbalance selects the model that is best at saying "Stage I". Class weights are applied as per-sample weights. Augmentation layers live *inside* the Keras model, which makes "augmentation on training data only" a structural guarantee rather than a convention: these layers are exact no-ops at inference. Colour augmentation (brightness, contrast) is deliberately aggressive, since Section IV-A shows colour is the dominant shortcut.

F. Evaluation

Primary metrics are quadratic-weighted κ (QWK) [19], mean absolute error in stages (MAE), macro-F1 and balanced accuracy. Supporting metrics are accuracy, adjacent (± 1 stage) accuracy, linear κ, per-stage specificity, one-vs-rest AUC and expected calibration error (ECE). Confidence is reported as the posterior mass of the *predicted* stage, $P(y = \hat{y})$, and as $P(|y - \hat{y}| \leq 1)$; each is calibrated against its own event. We fit a single temperature on internal validation data by minimising negative log-likelihood [20]. Confidence intervals are percentile bootstrap with 2,000 resamples.

G. Explainability

We compute Grad-CAM [21] from the final convolutional block. For ordinal heads we differentiate the *predicted-class posterior* $P(y = \hat{y})$ rather than the expected severity $\sum_k \hat{P}(y > k)$. This choice matters: for a confident Stage-I prediction every $\hat{P}(y > k) \approx 0$, so the severity gradient is degenerate and produces a diffuse map that is easily misread as "the model is attending to background". To quantify attention placement we report a *peripheral attention ratio*: the fraction of Grad-CAM mass falling outside a centred ellipse covering 55.7% of the frame. This is a geometric proxy, not scalp segmentation, and we report it as a rate rather than as proof.

IV. Results and Discussion

A. Audit Findings

Table I summarises the audit. The headline is that the 5,783 published files resolve to only **1,447 independent photographs**—a $4.0\times$ replication factor produced by colour-augmented copies that the export platform distributed across all three splits. Consequently 53.2% of the published test set and 53.8% of the validation set have a perceptual near-duplicate sitting in train, while byte-level deduplication reports a perfectly clean corpus.

Shortcut 1: the augmentation predicts the label. Fig. 1 shows the augmentation mix per stage. Stage I is 100% un-augmented and Stage II is 92.6% un-augmented, whereas Stages III–VII are each $\approx 80\%$ augmented. The binary feature "does this image carry a colour filter?" therefore separates

TABLE I
AUDIT OF THE PUBLISHED BENCHMARK

Property	Value
Published image files	5,783
Distinct source photographs	1,452
Independent groups (pHash ∪ source id)	**1,447**
Byte-identical duplicate groups	0
Test images with a near-duplicate in train	132 / 248 (53.2%)
Validation images with a near-duplicate in train	272 / 506 (53.8%)
Groups straddling two or more vendor splits	219
Groups with internally conflicting labels	6
Augmentation × stage, Cramér's V	0.390
Source family × stage, Cramér's V	0.432
Class imbalance, file level	7.2 : 1
Class imbalance, group level	**37.9 : 1**
Independent images in the rarest stage (VI)	19

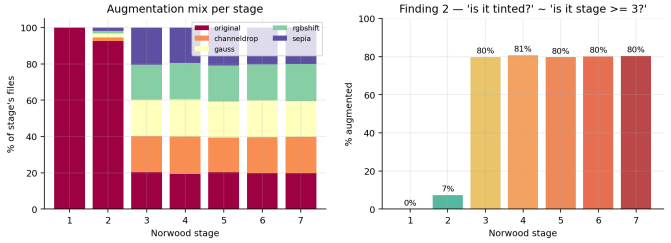


Fig. 1. Shortcut 1. Left: augmentation mix per stage. Right: percentage of each stage's files carrying a vendor colour augmentation. Stage I is entirely un-augmented while Stages III–VII sit near 80%, making the augmentation marker a near-perfect proxy for "Stage ≥ III".

{I,II} from {III–VII} almost perfectly ($\chi^2 = 3{,}527$, $V = 0.390$).

Shortcut 2: the source collection predicts the label. Filenames resolve to three collections. The `eu-<uuid>` and `exhibition_*` families contribute *zero* Stage-I images, yet together supply 58.0% of all Stage-VII files and 45.3% of Stage-VI files ($\chi^2 = 2{,}156$, $V = 0.432$). Camera, lighting, background and population are thus correlated with severity.

The shortcut floor. The colour-only control—no shape, no texture, no scalp structure—reaches QWK 0.722, accuracy 0.456 and 82.3% on the binary "Stage ≤2 vs. ≥3" task (chance 50.8%) on the published split. On our audited split the same features collapse to QWK 0.444 with accuracy 0.444, *below* the 0.499 majority-class baseline. We therefore take QWK 0.722 as the minimum bar on the published split: a reported number below it has not demonstrated that the model can see hair.

The effective sample is much smaller than it appears. Fig. 2 contrasts the file-level and group-level class distributions. Replication is strongly class-dependent—Stages III–VII appear 11.2–12.2× per independent photograph against 2.3× for Stage I—so the true imbalance is 37.9:1, not the 7.2:1 the file counts suggest, and Stage VI is represented by only **19 independent images in the entire corpus**.

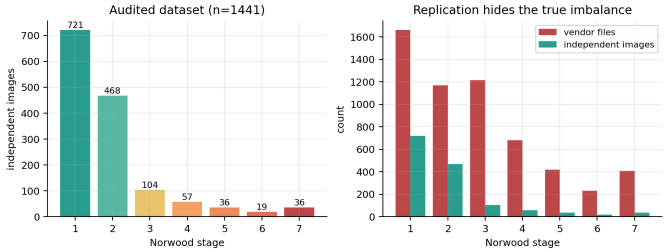


Fig. 2. Left: class distribution of the audited dataset (n=1,441 independent images). Right: file counts versus independent-image counts. Class-dependent replication conceals the true 37.9:1 imbalance.

TABLE II
CROSS-VALIDATED BENCHMARK, AUDITED PROTOCOL (POOLED OUT-OF-FOLD, 3 FOLDS)

Model	QWK	MAE	mF1	bAcc	Acc	±1
Majority class	0.000	0.906	0.095	0.143	0.499	0.824
Colour-only LR	0.368	1.230	0.249	0.317	0.421	0.714
Small CNN (scratch)	0.621	0.698	0.322	0.366	0.533	0.883
SVM (frozen RN50)	0.680	0.464	0.422	0.404	0.668	0.929
Ordinal SVM (frozen)	0.760	0.412	0.273	0.273	0.675	0.932
EfficientNet-B0 + CORN	0.737	0.672	0.337	0.361	0.455	0.905
ResNet-50 + CE (softmax)	0.869	0.340	0.520	0.531	0.714	0.960
ResNet-50 + CORAL	0.884	0.342	0.524	0.545	0.697	0.967
ResNet-50 + CORN	**0.900**	**0.283**	**0.580**	**0.590**	**0.747**	**0.975**

mF1 = macro-F1, bAcc = balanced accuracy, Acc = exact accuracy, ±1 = adjacent accuracy. Best value per column in bold.

TABLE III
ORDINAL HEADS VERSUS SOFTMAX, SAME TRUNK AND FOLDS

Head	ΔQWK	ΔMAE	ΔmF1	Δ±1
CE (softmax), reference	—	—	—	—
CORAL	+0.015	+0.003	+0.004	+0.007
CORN	**+0.031**	**−0.056**	**+0.060**	**+0.016**

B. Cross-Validated Benchmark

Table II reports pooled out-of-fold results over three folds of the development set (≈ 691 pooled predictions). Every deep model clears the audited colour floor of QWK 0.444, confirming that on the corrected protocol the networks are using information the colour control cannot access.

C. Does Ordinal Modelling Help?

This is the study's central question. Because all three heads sit on an identical trunk and see identical folds, the differences in Table III are attributable to the output formulation.

CORN improves on softmax on *every* metric, and is the best model on all six primary measures in Table II. The improvement is concentrated exactly where the ordinal hypothesis predicts: MAE falls by 0.056 stages and adjacent accuracy rises to 0.975, i.e. CORN makes fewer *large* mistakes rather than simply more correct ones. CORAL's gain over softmax is marginal (+0.015 QWK) and it is no better on MAE, which

TABLE IV
CALIBRATION AND POSTERIOR HEALTH (CROSS-VALIDATED)

Head	ECE	$ECE_{\pm 1}$	Valley rate	Rank viol.
CE (softmax)	0.046	0.020	0.0%	0.0%
CORAL	0.300	0.209	**48.3%**	0.0%
CORN	0.093	0.021	2.6%	0.0%

is consistent with the expressiveness cost of its single shared projection.

We state the strength of this evidence precisely. It is a pooled three-fold out-of-fold estimate without a paired per-sample significance test, so the $+0.031$ QWK margin is a consistent directional effect rather than a certified gap; the consistency—CORN leading on six of six metrics—is the more persuasive signal than any single margin.

D. Calibration and a CORAL Posterior Pathology

Table IV reports an unexpected and, to our knowledge, unreported failure mode. We define the *posterior valley rate* as the fraction of predictions for which the expected-value decode lands on a stage holding more than 0.25 less posterior mass than the arg-max stage—that is, the point prediction falls in a valley between two modes.

CORAL's posteriors are bimodal for almost half of all predictions, with mass piling at the extremes of the scale and the decoded stage falling between them; its ECE of 0.300 is roughly $6\times$ that of softmax. CORN, whose heads are free to specialise, shows a valley rate of only 2.6%. Rank-monotonicity violations are zero for all three heads—CORAL's learned biases did emerge ordered in practice, so its architectural guarantee was not needed—which strengthens rather than weakens the case for CORN: it obtains the same consistency *and* usable posteriors, without the constraint.

This matters clinically: a confidence score is only meaningful if it attaches to the stage actually reported. A system that decodes to Stage IV while placing its mass on Stages II and VII is not expressing uncertainty about Stage IV—it is hiding genuine bimodality behind a point estimate.

E. Locked Test Set

The final configuration (ResNet-50 + CORN) was retrained on the full development set with an internal validation carve-out for early stopping and temperature fitting ($T = 1.868$), then evaluated once on the locked test set. Table V reports the result with bootstrap intervals; Fig. 3 shows the confusion structure.

The error structure is the result we set out to obtain: 97.6% of predictions land within one stage, 2.4% are two or more stages off, and a single case (0.3%) is off by more than two. Per-stage, the model is strongest at the extremes (Stage I recall 0.792, Stage VII recall 0.875, specificity 0.996) and weakest in the transition zone (Stage III recall 0.429, n=21), which mirrors where human raters disagree. Stage VI has only three test images and its recall of 0.333 carries a confidence interval spanning most of the unit interval; we report it for completeness and decline to interpret it.

TABLE V
LOCKED TEST SET, EVALUATED ONCE ($n = 289$)

Metric	Value	95% CI
Quadratic-weighted κ	0.891	[0.832, 0.927]
MAE (stages)	0.311	[0.253, 0.377]
Macro-F1	0.641	[0.533, 0.738]
Balanced accuracy	0.652	[0.546, 0.775]
Exact accuracy	0.720	[0.671, 0.772]
Adjacent (± 1) accuracy	0.976	[0.955, 0.993]
Linear-weighted κ	0.748	—
Unweighted κ	0.565	—
ECE (exact)	0.097	—
ECE (± 1 stage)	0.019	—

Interpreting QWK against the human ceiling. QWK 0.891 exceeds the ICC 0.63–0.68 of expert raters [4], which would be an immediate red flag if the two numbers were comparable. They are not. We verified that the skewed class marginal is not responsible: on a class-balanced resample of the test set QWK is 0.910 [0.875, 0.940], slightly *higher*. The explanation is the weighting. QWK discounts adjacent disagreements quadratically, and our model's errors are almost all adjacent. The directly comparable quantity for exact agreement is unweighted $\kappa = 0.565$, with linear-weighted $\kappa = 0.748$ in between. Read this way the result is unsurprising and honest: exact agreement is moderate and broadly in line with human-level performance on an unreliable scale, while the high QWK reports the specific, clinically desirable property that the model rarely misses by much. **We therefore caution against reporting QWK alone on this task.**

Confidence is usable for triage. Calibration on the clinically meaningful event is good ($ECE_{\pm 1} = 0.019$), and confidence separates correct from incorrect predictions (mean 0.743 versus 0.562). Thresholding at $P(y = \hat{y}) \geq 0.7$ retains 49% of cases at 90.8% exact accuracy and 100% adjacent accuracy; at ≥ 0.8 it retains 31% at 95.6% accuracy. A deferral policy of this kind is the realistic deployment shape for a second-opinion tool.

F. Robustness to Residual Similarity

Grouping removes copies of the same *photograph*. It cannot remove a second photograph of the same *patient* from another session, because the dataset carries no patient identifiers. We bound this residual risk by re-scoring on progressively cleaner subsets of the test set, excluding images whose nearest development-set neighbour lies within a given perceptual distance (Table VI).

QWK is essentially invariant across the sweep (range 0.889–0.894) even as the test set shrinks by 39%. Exact accuracy declines modestly from 0.720 to 0.670, indicating that a small part of the exact-match performance—but not the ordinal agreement—is attributable to visually similar sessions. **Residual patient-level leakage does not drive our result.**

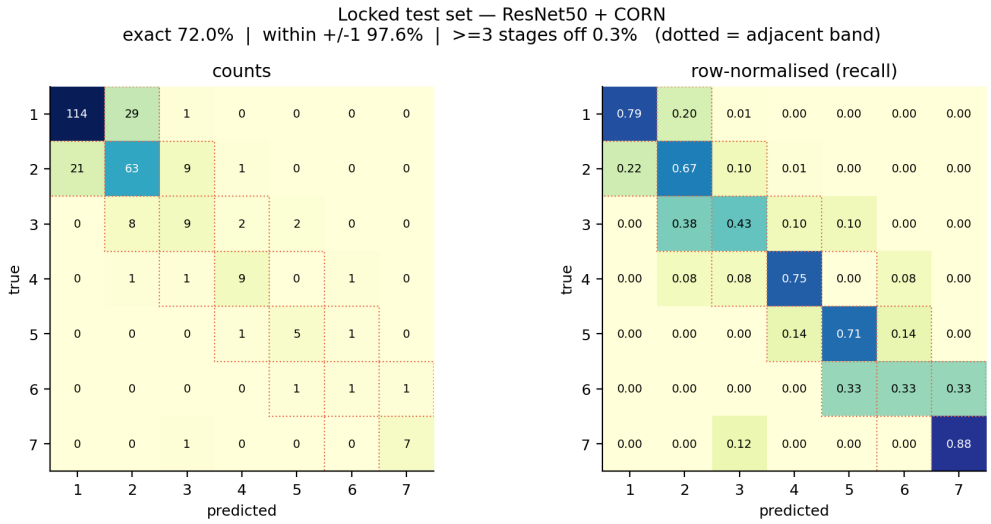


Fig. 3. Locked-test confusion matrix (counts, left; row-normalised recall, right). Error mass is almost entirely confined to the adjacent band (dotted outline). Only 2.4% of predictions are two or more stages off and 0.3% are three or more. The model is strongest at the scale extremes and weakest in the Stage III transition zone, mirroring where human raters disagree.

TABLE VI
SENSITIVITY TO RESIDUAL NEAR-DUPLICATE SIMILARITY

Test subset	n	**QWK**	**MAE**	**Acc.**
All test images	289	0.891	0.311	0.720
No neighbour within 6 bits	282	0.892	0.312	0.720
No neighbour within 8 bits	275	0.894	0.309	0.724
No neighbour within 10 bits	238	0.891	0.340	0.697
No neighbour within 12 bits	176	0.889	0.369	0.670

TABLE VII
LEAKAGE ABLATION: IDENTICAL MODEL, RECIPE AND SEED

Metric	**Vendor split**	**Audited split**	**Inflation**
QWK	0.972	0.891	+0.081
Accuracy	0.871	0.720	+0.151
Macro-F1	0.893	0.641	+0.253
MAE (stages)	0.145	0.311	−0.166
Colour-only control	*0.722*	*0.444*	+0.278

G. Explainability

Fig. 4 shows Grad-CAM overlays for correct and incorrect locked-test predictions. For advanced stages the evidence is clinically plausible: attention concentrates on the vertex and crown, the regions that define Stages IV–VII. For early stages the maps are more diffuse and occasionally drift to image borders, which is where the model is also least accurate.

Quantitatively, the peripheral attention ratio sits well *below* the 0.443 value a uniform map would produce: per-stage medians run from 0.19 at Stage I to 0.42 at Stage III (Fig. 5), i.e. attention is predominantly on-scalp. The ordering is informative: attention is least peripheral at Stage I and most peripheral at Stage III, the stage the model also classifies worst (recall 0.429)—where it is least accurate it is also least focused.

We report a methodological caution here, because we initially observed the opposite. In development runs using the expected-severity target $\sum_k \hat{P}(y > k)$, peripheral ratio fell monotonically across stages, from roughly 0.8 at Stage I to below 0.3 at Stage VII—a pattern that invites the conclusion that the network reads background to recognise Stage I. It does not follow. For a confident Stage-I prediction the severity quantity is saturated near zero, its gradient is numerically degenerate, and the resulting map is close to noise. Switching to the predicted-class posterior removes the artefact entirely. **The choice of Grad-CAM target is not cosmetic for ordinal heads, and a stage-dependent diffuseness trend should be treated as a null result until the target is checked.**

H. The Leakage Ablation

Table VII and Fig. 6 present the study's practical headline. The *same architecture, the same hyper-parameters and the same random seed* were trained and evaluated twice, differing only in the data protocol.

The unaudited benchmark inflates accuracy by 15.1 points, QWK by 0.081, and macro-F1 by 0.253. Macro-F1 is the most distorted because leakage disproportionately benefits the rare advanced stages, which are precisely the classes that are most heavily replicated. Reported on the published split, this model would appear to achieve 87.1% accuracy and QWK 0.972—numbers comparable to those in the recent literature, and, on this evidence, not measuring generalisation.

We note one further diagnostic. We attempted to evaluate the vendor-trained model on our own group-disjoint test set, but 73.7% of that set is already in the vendor's training split and 89.6% is in its train-or-validation split, leaving only 30 genuinely unseen groups, all from Stages I–II. **No fair test**



Fig. 4. Grad-CAM on the locked test set, differentiating the predicted-class posterior. For Stages IV–VII attention localises on the vertex and crown. Errors on early stages (e.g. true Stage I predicted Stage III, second column) show attention drifting toward image borders. The true Stage-II / predicted Stage-IV case (fourth column) is a female scalp with a central parting, a presentation the Norwood scale does not describe.

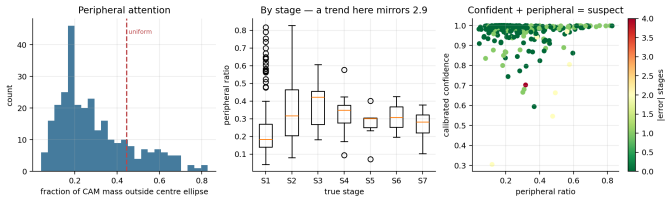


Fig. 5. Peripheral attention analysis. Left: the distribution sits below the 0.443 uniform reference (dashed), i.e. attention is concentrated inside the central scalp region. Centre: by true stage, with no strong monotone trend once the correct Grad-CAM target is used. Right: confidence versus peripheral ratio, coloured by stage error.



Fig. 6. The same model scored under both protocols. The dotted line marks the colour-only shortcut floor on the vendor split.

set for the vendor-trained model can be constructed from this dataset at all. That impossibility is arguably a cleaner statement of the severity of the leakage than any number we could have reported alongside it.

I. Limitations

Sample size. The corpus holds 1,441 independent images; Stage VI is represented by 19 in total and three in the locked test set. Per-class figures for Stages V–VII should be read from the pooled cross-validated results, not from the single hold-out.

No patient identifiers. Independence is established at the photograph level, not the patient level. Table VI bounds the consequence but cannot eliminate it.

Irreversible pre-processing. Images arrived stretch-resized to 640×640 with EXIF stripped, so original framing and aspect ratio are unrecoverable.

Single view. The corpus is predominantly top-down and vertex-oblique. The Norwood scale is defined on frontal *and* vertex patterns, and frontal hairline recession is frequently not visible.

Population and scope. One dominant clinical source, no demographic metadata, so fairness across skin tone, age or sex cannot be assessed. Some images are evidently female scalps (Fig. 4), which the Norwood scale does not cover. The locked test set is $282/289$ from a single collection, so cross-collection generalisation is untested; conversely this means the collection confound cannot explain the test-set result.

Statistical strength of the head comparison. As noted, the CORN-versus-softmax margin rests on a pooled three-fold estimate without a paired significance test.

Evaluation discipline. The locked split is SHA-256 identical across a preliminary pipeline-verification run and the reported run. No architecture, hyper-parameter or threshold was chosen using test data—model selection used development-set cross-validation throughout—but we record the fact for transparency.

V. Conclusion

We set out to test whether treating Norwood–Hamilton staging as an ordinal problem produces more clinically meaningful results than flat classification, and to do so on a benchmark whose integrity we had verified rather than assumed. Both halves produced results.

The audit was not a formality. The most widely available public AGA dataset contains 53.2% test-set leakage that byte-level deduplication cannot see, two label-confounded non-clinical artefacts, and an effective sample size four times smaller than its file count. A model using only 28 colour statistics scores QWK 0.722 on it. Under a controlled ablation, the unaudited split inflates accuracy by 15.1 points and macro-F1 by 0.253.

On the corrected protocol the ordinal hypothesis is supported. CORN leads on all six primary metrics, improving over softmax by $+0.031$ QWK and -0.056 stages of MAE, with the gain concentrated in the reduction of large errors. On a locked test set it reaches QWK 0.891 [0.832, 0.927] with 97.6% of predictions within one stage and only 0.3% off by three or more, calibrated to ECE 0.019 on the ± 1 event and robust to residual near-duplicate similarity. We also document a posterior pathology in CORAL—bimodal outputs in 48.3% of predictions—that makes its confidence scores unusable despite competitive point accuracy.

Two recommendations follow for the field. First, any result on a public AGA benchmark should be accompanied by a group-disjoint split and a trivial-feature control; the cost is an afternoon and the alternative is reporting an artefact. Second, QWK should not be reported alone on this task: paired with unweighted κ it tells an honest story, and alone it flatters.

Future work needs better data more than better architectures: patient identifiers for genuine subject-level splitting, a second independently collected cohort for external validation, and paired frontal and vertex views so a model can see what the Norwood scale actually describes. We release the audit code, the locked split manifest and a model card to make each directly buildable.

This work is a research prototype. It is not a diagnostic device, makes no treatment recommendation, and is not a substitute for clinical judgement.

Reproducibility

Single end-to-end notebook run; seed 1337; TensorFlow/Keras. Locked split manifest SHA-256:

`dc94878c954130dc3aa7140dae1043de`
`7f5660f27d41f910b7ec846a3f51e193`

Released artefacts: audit and split manifests, per-image test predictions with calibrated confidences and nearest-neighbour distances, the cross-validated benchmark, the leakage ablation, the residual-similarity table, and a model card.

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